Beating the Berkeley Baseline in Simulation: Robust PPO Locomotion for a Low-Cost Humanoid Biped

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Abstract

We present a PPO training recipe for robust locomotion on the 12-DoF Berkeley Humanoid Lite biped, developed and validated through a 230-run, ∼268-hour campaign: lowered command-tracking weights with three added regularization terms, a left/right-symmetrized asset family, push randomization, and a DeepMimic-style motion-imitation objective, trained with RSL-RL in Isaac Lab. In a controlled comparison of 15 recipe variants (byte-for-byte identical in network architecture and deploy configuration, so every difference is attributable to a named ingredient), the recipe's best variant, `R40_FullSymUSD`, is the only policy of the campaign to fall zero times across all 73 MuJoCo deep-evaluation scenarios while clearing a both-engine robustness bar (one of 7 of the 230 logged runs), and it ranks above the upstream Berkeley reference on cross-engine robustness within our audit. Validation in a second, independent physics engine is integral to the recipe: a three-layer protocol (a 7-scenario MuJoCo plus 6-scenario Isaac Sim gate, a 73-scenario MuJoCo deep evaluation, and a 336-push robustness matrix in both engines) shows that the highest-reward variants overfit the training engine, the lowest-tracking-error runs place bottom-two in Isaac, and one run drifts +48% in tracking error between its peak and final checkpoints, so no single-engine score identifies the robust policy. The strict zero-fall envelope (0 falls in *both* engines) holds to ≤ 20 N; the ≤ 40 N range is operator-safe (1/84 MuJoCo, 0 Isaac); push survival is 90.5% in MuJoCo and 55.1% in Isaac. All results are in simulation: the robot is calibrated and the ROS 2 stack drives its motors; closed-loop walking on hardware is the next milestone.

1 Introduction

Learning-based control has become the dominant approach to humanoid locomotion: a policy is trained in large-scale parallel simulation and transferred to hardware [2, 10, 5]. Our platform is the *Berkeley Humanoid Lite* [6], a sub-$5000, 3D-printed, open-source 12-degree-of-freedom (DoF) biped from UC Berkeley, and we target locomotion that is *robust*: stable under external disturbance and beyond the exact conditions of training, rather than merely maximal on a training score. We build *on* the platform's upstream pipeline: a Proximal Policy Optimization (PPO) agent [1] is trained in NVIDIA Isaac Lab [7] via the RSL-RL "massively parallel" deep-RL implementation [8], exported to ONNX (Open Neural Network Exchange), validated cross-engine in MuJoCo (sim2sim), and deployed to the robot. We change only the training recipe and how policies are selected from it. Section 2 situates this within prior work on parallel-simulation locomotion, sim-to-real transfer, and cross-engine evaluation, and marks where our contribution departs from it.

The campaign and the recipe. The recipe was developed over a 230-run, eight-week campaign (Section 4.3) that systematically varied reward shaping, asset symmetry, and the disturbance curriculum. Its ingredients are explicit: lowered command-tracking weights paired with three added penalty terms (the trunk of the recipe), a left/right-symmetrized USD asset family, a `push_robot` disturbance knob, two fine-tuning families (one adding a DeepMimic-style motion-imitation objective) and a tightened action clamp, each itemized in Section 4. Its controlled evaluation stage is **15 policies that are byte-for-byte identical in network architecture and deploy configuration**, holding the PPO and network hyperparameters fixed and varying *only* the recipe (Table 1 the held-fixed-versus-varied diff, Table 2 the recipe glossary), so every downstream difference is attributable to a small, explicit set of recipe changes rather than to a confounded hyperparameter.

Cross-engine validation. Our instrument for judging robustness is a three-layer evaluation that scores every candidate in a second, independent physics engine: a 7-scenario MuJoCo and 6-scenario Isaac Sim gate, a 73-scenario MuJoCo deep evaluation, and a 336-push robustness matrix in *both* engines (Section 6). Surviving a simulator the policy was never trained in distinguishes genuine robustness from overfitting in a way the training metrics alone cannot certify, and on this platform that distinction is decisive. The highest-reward PPO variants overfit the engine they were trained in: the lowest-tracking-error April runs land bottom-two in Isaac, the best MuJoCo-tracking policy is not zero-fall, and reading one run at its peak versus final checkpoint shifts err_xy by +48% (Section 6). `R40_FullSymUSD (rank01)` is the only one of the 15 recipe variants with 0/73 deep-eval falls (flag: base-height drop > 0.15 m or pitch/roll $> 60°$) and, with rank02, one of two variants, and one of 7 of the 230 trained runs, to clear the both-engine bar (0 falls in both engines, top-third Isaac composite) that the overfit high-reward runs fail. The robustness study additionally proposes the tied `R40_V10SymUSD (rank03)` as the live push-

deploy variant, on its symmetric left/right push survival (0.0 versus 1.2 pp), not on survival count.

Two boundaries hold throughout. The comparison to the upstream Berkeley reference is a within-audit cross-engine robustness result: the reference enters the comparison as an unaudited control (Section 6.2), not a published-benchmark head-to-head. And every number is a simulation number, pinned to MuJoCo or Isaac Sim: the robot is calibrated and the deploy stack drives its motors (Section 5), and closed-loop walking on hardware is the next milestone. Figure 1 shows the platform.

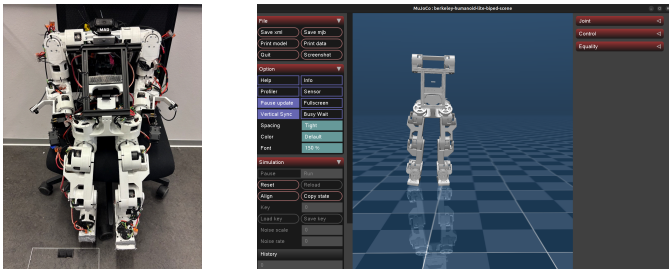


Figure 1. Left: our Berkeley Humanoid Lite [6] build; the biped task controls its 12-DoF lower body. Right: the same robot in the MuJoCo sim2sim scene, the second, independent physics engine of our cross-engine protocol: policies are trained in Isaac Lab and replayed here, so that surviving an engine the policy never saw separates genuine robustness from overfitting to the training simulator. Every audited policy is scored in this scene. *(sources: hardware photo 2026-06; figures/mujoco_sim2sim_biped.png, scene from evidence/sim2real.md, §(b), `playback/locomotion/mujoco.py`.)*

Contributions. We contribute, first, a concrete and reproducible PPO training recipe for robust biped locomotion, anchored by its controlled 15-policy comparison that ties each gain to an explicit ingredient. Second, we contribute a cross-engine validation protocol that scores candidates in a held-out physics engine and, in doing so, exposes training-engine overfitting that single-simulator metrics cannot detect. Third, we contribute a catalog of negative results from the broader campaign (AMP, warm-start transfer, a gait clock, and no-strafe gait shaping) that delimits the design space around the recipe.

The remainder proceeds as follows. Section 2 positions the work against prior parallel-simulation locomotion, sim-to-real, and cross-engine evaluation; Section 3 reviews the RL/PPO background and training stack; Section 4 details the controlled recipe diff and the broader campaign; Section 5 states the cross-engine protocol and hardware boundary; Section 6 reports the cross-engine-selected 0/73 policy, the overfitting evidence, the peak-versus-final regression, and the cross-engine push study; and Section 7 and Section 8 conclude.

2 Related Work

Massively parallel reinforcement learning has made training a legged-locomotion policy fast. Rudin et al. [8], whose RSL-RL implementation we use, brought wall-clock training for legged robots down to a minutes-scale regime by running thousands of environments on a single GPU; Isaac Lab and its Orbit predecessor [7] supply the simulation infrastructure that makes this regime routine. Radosavovic et al. [5] carried the train-in-simulation paradigm to a real humanoid at scale, and Seo et al. [9] compressed the sim-to-real recipe further to a fifteen-minute budget. Replaying a trained policy in a second physics engine is common engineering practice within this line: it appears as a validation step in fast sim-to-real recipes [9] and in the Berkeley Humanoid Lite pipeline [6], and the humanoid survey of Gu et al. [10] documents the sim-to-real gap such replay is meant to probe. That practice typically treats the second engine as a sanity check on a single chosen policy. This machinery supplies our training and validation substrate; our contribution is a concrete recipe developed on top of it; the downstream question of which policy to deploy (a choice the training pipeline does not itself resolve) is handled by treating the second engine as a formal, multi-scenario instrument applied across a controlled fifteen-policy zoo, where it rejects policies that single-engine metrics accept.

A second line concerns crossing the gap between the training simulator and the world. Tobin et al. [3] established domain randomization, perturbing simulator dynamics so a single policy is robust across the induced distribution, and RMA [4] paired randomization with an online adaptation module for legged robots. We inherit an upstream randomization recipe and add a single push-disturbance knob to it, but we propose no new transfer mechanism. Our recipe instead composes known ingredients (domain randomization, a left/right-symmetrized asset family, and motion imitation) and documents which of them mattered on this platform.

Motion imitation supplies one of our recipe families. DeepMimic [11] introduced a phase-indexed pose-tracking reward, and AMP [12] replaced the hand-specified phase reference with an adversarial style discriminator. One of our fine-tuning families uses a DeepMimic-style reward and produces the highest Isaac composite of the audited set (rank02); we also implemented an AMP discriminator over RSL-RL, but it was built and validated only to an integration smoke test, not carried to a full training run (Section 7). The other side of this development is selecting a policy or checkpoint for deployment. That proxy metrics (training reward, or single-simulator tracking error) can mis-rank policies for deployment robustness is, to our knowledge, not systematically examined for humanoid locomotion on a controlled multi-policy set. Our campaign surfaces this as a supporting finding: a hazard our validation protocol exposed and that recipe developers should expect, including a checkpoint-level instance in which tracking error drifts +48% between the peak and final checkpoints of a single run, so that the criterion used to export a policy itself changes which checkpoint is chosen.

3 Background: Reinforcement Learning and PPO

We train the walking controller with reinforcement learning (RL), and specifically with Proximal Policy Optimization (PPO). This section fixes the notation and the two objectives we rely on in Section 4; for a fuller treatment we follow Murphy's overview [2].

3.1 RL primer

We model locomotion as a Markov decision process: at step t the agent observes state s_t, picks action $a_t \sim \pi(\cdot \mid s_t)$, transitions to s_{t+1}, and receives reward r_t [2]. On the robot s_t is the 45-dimensional observation (velocity command, base angular velocity, projected gravity, joint

positions and velocities, previous action) and a_t is the 12 joint-position targets, one per actuated degree of freedom (DoF). The discounted return and value function (1) are

$$G_t = \sum_{k=0}^{T-1-t} \gamma^k \, r_{t+k}, \qquad V_\pi(s_t) = \mathbb{E}[\, G_t \mid \pi \,], \tag{1}$$

with $\gamma \in [0, 1]$; the optimal policy maximizes $\mathbb{E}_{p_0(s_0)}[V_\pi(s_0)]$, the "maximize accumulated reward" objective the walking policy is trained against [2]. The methods here are actor–critic: a deployed actor π_θ and a training-only critic V_ϕ that reduces gradient variance via generalized advantage estimation (GAE) [1]. Both are $[256, 128, 128]$-ELU multilayer perceptrons; the critic additionally sees base linear velocity (48-d input versus the actor's 45) and is discarded at deployment.

3.2 Policy gradient and PPO

Policy-gradient methods optimize θ by stochastic gradient ascent on (2):

$$\hat{g} = \hat{\mathbb{E}}_t \Big[\nabla_\theta \log \pi_\theta(a_t \mid s_t) \, \hat{A}_t \Big], \tag{2}$$

where $\hat{A}_t$ estimates the advantage and $\hat{\mathbb{E}}_t[\cdot]$ averages over a batch [1]. Running multiple epochs on this surrogate over one trajectory causes destructively large updates [1], motivating clipping: PPO uses the probability ratio (3) between the current and data-collecting policy,

$$r_t(\theta) = \frac{\pi_\theta(a_t \mid s_t)}{\pi_{\theta_{\text{old}}}(a_t \mid s_t)}, \qquad r_t(\theta_{\text{old}}) = 1, \tag{3}$$

and maximizes the clipped surrogate [1]

$$L^{CLIP}(\theta) = \hat{\mathbb{E}}_t \big[\min \big(r_t(\theta) \, \hat{A}_t, \; \text{clip}(r_t(\theta), \, 1-\epsilon, \, 1+\epsilon) \, \hat{A}_t \big) \big], \tag{4}$$

with clip hyperparameter $\epsilon = 0.2$. The minimum of the clipped and unclipped terms makes L^{CLIP} a pessimistic lower bound that keeps each update in a small trust region near the old policy [1]; the full objective adds a squared value-function error and an entropy bonus. The advantages $\hat{A}_t$ in (4) use truncated GAE, given in (5):

$$\hat{A}_t = \sum_{l=0}^{T-t-1} (\gamma\lambda)^l \, \delta_{t+l}, \qquad \delta_t = r_t + \gamma V_\phi(s_{t+1}) - V_\phi(s_t), \tag{5}$$

with $\lambda = 0.95$ [1]: the exponentially-weighted sum trades bias against variance ($\lambda = 1$ the full Monte-Carlo return, $\lambda = 0$ the one-step TD residual), used exactly as in [1]. The PPO hyperparameters ($\epsilon = 0.2$, entropy 0.008, adaptive lr 10^{-3}, $\gamma = 0.99$, $\lambda = 0.95$, 5 epochs / 4 minibatches, 24 steps/env, the $[256, 128, 128]$ ELU MLP) are held fixed for every run, read directly from the RSL-RL configuration (`.../config/biped/agents/rsl_rl_ppo_cfg.py`, cross-checked in `evidence/background-citations.md`, §2).

3.3 Training stack: RSL-RL inside Isaac Lab

The optimizer is RSL-RL, ETH Zürich's on-policy RL library whose canonical reference is the massively-parallel-on-GPU locomotion work of Rudin et al. [8], plugged into NVIDIA Isaac Lab (the renamed successor of NVIDIA *Orbit* [7]); we train with $\sim$ 2048–4096 parallel environments. RSL-RL rescales the learning rate each update to hold the measured policy KL near `desired_kl`$= 0.01$, its analogue of PPO's adaptive-KL idea [1]. The robot and its pipeline are the Berkeley Humanoid Lite project [6], within the humanoid-locomotion landscape surveyed in [10]; our task, reward design, and safety changes are in Section 4.

Figure 2 shows the training histories of three audited recipes and the highest-reward April run (`bigger_envs`), retrieved via the W&B API. The two panels serve distinct purposes: the per-run mean episode reward (left) shows within-run convergence but is *not* comparable across recipes, because each recipe changes the reward composition: `bigger_envs` relaxes regularization terms while `AblImit_FT` adds an imitation reward, so a higher mean reward in one run does not indicate a better policy. The command-tracking error `err_xy` (right) is a physics quantity independent of reward shaping and *is* directly comparable across runs.

3.4 Strategies adopted from the literature

The training recipe is not invented from scratch: each concrete ingredient maps to a specific technique from the literature, traced in prose below. Every mapping is traceable to `evidence/background-citations.md` (the papers read) and `evidence/baseline-diff.md` (the recipe ingredients); items with no citation are recipe choices grounded in those notes rather than a single paper.

The *optimizer* is Proximal Policy Optimization [1]: the clipped-surrogate objective of (4) with truncated GAE and value/entropy terms, used exactly as published behind all 15 runs, realized through RSL-RL, whose canonical massively-parallel-on-GPU reference is Rudin et al. [8] (the $\sim$ 2048–4096 parallel-environment regime and adaptive-KL schedule come from that line), with NVIDIA Isaac Lab / Orbit [7] as the simulation framework. *Domain randomization* of ground friction, base mass, actuator stiffness/damping, default joint positions, and reset kicks (inherited from upstream and augmented with the `push_robot` knob) is the randomization-for-transfer strategy whose canonical reference here is RMA [4]; we adopt the idea without RMA's explicit online adaptation module.

Symmetry augmentation is enforced at the asset level: the symmetrized USD family (`SymUSD`, `FullSymUSD`, `V10SymUSD`) mirror-averages base and leg inertia (and, for `V10SymUSD`, per-body centre of mass), removing the CAD-export asymmetry that otherwise induces a $\approx 0.22°$/s leftward drift, and is the recipe axis that selects the deployed policy (Section 4). A mirror-symmetry loss (`Loss/symmetry`) was logged during sym-tagged runs, but its gradient was disabled (`use_mirror_loss=False`) in the audited biped runs: the deployed symmetry is structural, not a trained loss term. The training also uses an *asymmetric actor–critic* [1, 2]: the critic sees a privileged observation (base linear velocity, 48-d versus the actor's 45-d) and is discarded at deployment, reducing gradient variance without leaking into the deployed policy. The overall train-in-parallel-sim and transfer-to-hardware paradigm is that demonstrated by Radosavovic et al. [5], within the fast-sim2real and survey context [9, 10]. One technique we deliberately did *not* deploy: a sin/cos gait-

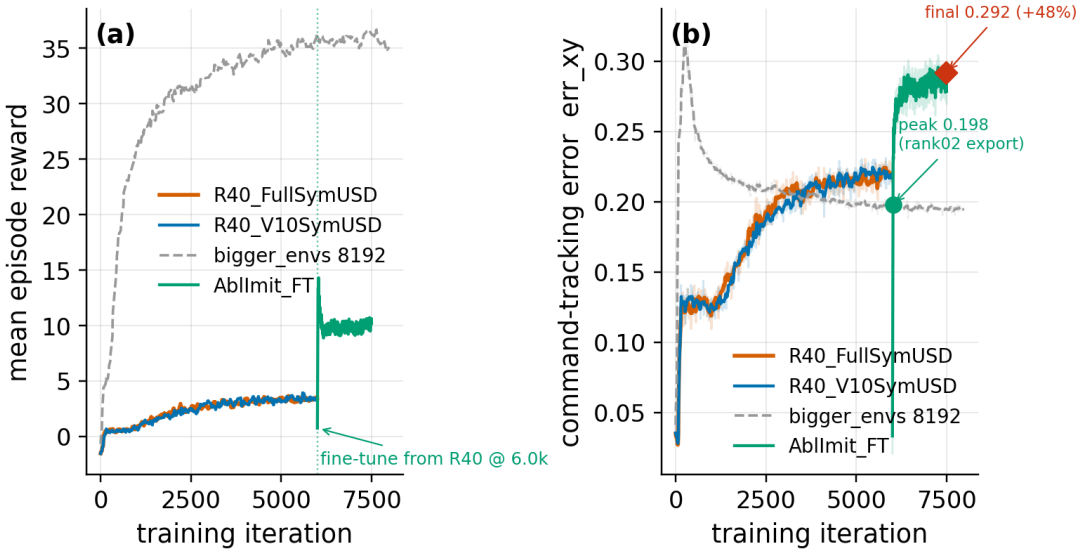


Figure 2. Training histories for three audited biped recipes and the highest-reward April run (W&B runs fefxxxr7, c46z0ixv, 1uwo7px2, aqe6jlq5). **Left: per-run mean episode reward**, *not* comparable across runs: `bigger_envs` uses relaxed regularization (inflating absolute reward) and `AblImit_FT` adds an imitation term absent elsewhere, so cross-recipe reward ranking is not a selection signal (the panel shows only within-run convergence). **Right: command-tracking error `err_xy`**, independent of reward composition and directly comparable. `AblImit_FT` (run aqe6jlq5, history iterations 5999–7498) reaches its `err_xy` minimum of 0.198 at iteration 6013 (the rank02 export), then drifts to 0.292 at iteration 7488 (+48%, the rank07 checkpoint). `bigger_envs` reaches the lowest training `err_xy` of all runs yet places at the bottom of the Isaac composite (Section 6.3), illustrating reward-shaping–induced over-optimism. (source: W&B API histories, alex-huou/mina runs fefxxxr7, c46z0ixv, 1uwo7px2, aqe6jlq5; `figures/data/`.)

phase clock (raising the observation to 77 dimensions) was explored only in later standing/balance runtimes, while the deployed 12-DoF biped keeps the upstream 45-d observation with *no* phase clock.

Motion imitation. Two recipe families add an explicit imitation objective from the character-animation literature. The `AblImit_FT` family adds a DeepMimic-style [11] phase-indexed pose-tracking reward, $r_{\text{imit}} = \exp(-\sigma\|q_{\text{ref}}(\phi) - q\|^2)$, with $q_{\text{ref}}(\phi)$ a per-joint leg target read from a reference walking clip at gait phase ϕ (command-gated and multi-clip walk/turn variants; clips retargeted from motion capture and upstream-policy rollouts). It produces `rank02`, the highest Isaac composite of the audited set (0.502) and, with symmetrized-asset `rank01`, one of the two controlled-set variants, and one of the seven policies overall, to reach Tier-1 (Section 6). We also implemented an adversarial-imitation (AMP [12]) discriminator from scratch over RSL-RL, but validated it only to an integration smoke test, not a full training run; the phase-indexed reward is the imitation route that reached the audited set (Section 7).

4 Developing the Recipe: Platform, Campaign, and the Final Configuration

This section fixes the platform and learning problem, the training infrastructure, the eight-week campaign that generated the candidate pool, and (as a line-level diff against the public Berkeley Humanoid Lite baseline) exactly what the final recipe holds fixed and changes. Because the 15 audited policies are identical in architecture and deploy configuration and differ only in the training recipe, every effect measured in Section 6 is attributable to that recipe and nothing else, which licenses reading the cross-engine divergences as a property of the recipe rather than of confounded hyperparameters. The recipe is not limited to reward weights: it spans domain randomization, a symmetrized asset family, two fine-tuning families, and a DeepMimic-style motion-imitation objective (Section 3.4). Architecture and observation changes and an adversarial-imitation network were explored outside this audited set and reported as negative results (Section 7). We keep the recipe-family naming used in the artifacts (the `R40` trunk, the symmetrized asset family, the fine-tune (FT) families `AblImit_FT` / `Gait_FT`); the forward glossary is Table 2.

4.1 Platform and observation/action space

The platform is the Berkeley Humanoid Lite [6]; all policies use the biped task (`Velocity-Berkeley-Humanoid-Lite-Biped-v0`), a 12-DoF lower body of six leg joints per side actuated as joint-position targets. The 45-d observation (velocity command, base angular velocity, projected gravity, joint positions/velocities, previous action) and 12-d action (action scale 0.25 on the default offset), the PD gains (k_p=20, k_d=2, 6 N m limit), and the training command ranges ($v_x = \pm 0.5$, $v_y = \pm 0.25$ m/s, $\omega_z = \pm 1.0$ rad/s, heading enabled) are all inherited unchanged from upstream (Table 1); the MDP and policy-gradient background are in Section 3.

4.2 Training infrastructure

We train with PPO [1] via the RSL-RL implementation in NVIDIA Isaac Lab [8, 7], which runs thousands of robot copies in parallel on one GPU. The PPO and network hyperparameters (the [256, 128, 128] ELU actor/-critic MLP and the clip/entropy/learning-rate/GAE settings of Section 3) are held fixed across all 15 runs. Runs span a local workstation (2048 parallel envs) and a SLURM/L40S cluster (4096, a few at 8192), all seeded *s*42 (encoded in the run name; rank08 and rank15 carry no seed suffix) for 6000 iterations (to 12000 on larger runs). Training-scale fields are read from run-name strings, not a logged config column (all 230 run-database

rows have empty `cfg/num_envs/max_iterations/seed`); with no per-iteration log retained locally for any final 6000-iteration policy, the converged episode length (≈ 340 steps) is taken from the audited checkpoints of Section 6 and the convergence signature from Figure 2 (Section 3).

4.3 The experimental campaign

The 15 audited policies are the final stage of a 230-run campaign spanning 2026-03-22 to 2026-05-16 (approximately eight weeks), which proceeded in five recognizable phases. (1) A baseline and domain-randomization exploration phase (the largest by run count) on the full humanoid model and early biped configurations, which established that relaxing regularization and scaling to 8192 parallel environments inflate training reward without certifying cross-engine robustness (the overfit pattern of Section 6.3). (2) A reward-crafting cohort of single-term ablations on the biped (the R0–R31 series in `experiment_name`), isolating which reward terms are necessary for gait quality. (3) The symmetrized-asset ladder and fine-tune families (`R40` → `SymUSD` → `FullSymUSD`/`V10SymUSD` plus the imitation and gait fine-tune cohorts) that produced the 15 audited policies. (4) A raised-shoulder transfer study (humanoid-biped `--resume` warm-starts to a taller CoM configuration): all 16 attempts failed to retain the source-task gait, with the mechanism reported in Section 7. (5) An adversarial-imitation (AMP) track of five cluster runs, none carried to a full training run, yielding an implementation validated to smoke-test level but no audited policy.

The 230 runs partition into mutually exclusive families by `experiment_name` (priority-order matching): AMP (5 runs, 0 finished), the R40 / symmetrized-asset ladder (13 runs, 9 finished), the gait / imitation fine-tune families (11 runs, 8 finished), other biped configurations (75 runs, 31 finished, covering the reward-crafting R0–R31 cohort, V31 standing/balance variants, and vanilla/transfer setups), and the humanoid stream (126 runs, 62 finished, which includes the raised-shoulder `MinaRaised` transfer study, whose runs carry humanoid experiment names): 110 of 230 runs finished overall. The humanoid stream accounts for more than half the runs and more than half the logged hours, reflecting the early exploration phase, while the symmetrized-asset ladder and fine-tune families together total 24 runs and 27.9 h, the smallest fraction of the campaign, yet the source of all audited policies. Logged training wall-clock is ≈268 h across all 230 runs (∼97 h on the biped stream), and includes partial time of crashed and failed runs (source: figures/data/campaign_stats.md; runs.csv.).

The breadth of this search is what makes the selection problem substantive: with 230 runs and a leaderboard, some single-number ranking will be applied unless a more principled procedure exists, which motivates the cross-engine audit protocol of Section 6.1.

4.4 What we changed versus the Berkeley baseline

Table 1 is the full line-level diff. The PPO and network hyperparameters, the observation and action space, the PD gains, the control rates and the core domain randomization (DR) were all left unchanged; deliberate variation lives in five places: reward weights, a symmetrized USD asset family, a `push_robot` disturbance knob, two FT families, and the action clamp. Three need qualification. First, the unambiguous reward deltas are the lowered tracking weights ($+2.0/+1.0 \rightarrow +1.5/+0.75$) and three *added* terms (`dof_vel_l2` -0.001, `action_l2` -0.001, `joint_deviation_legs` -0.5); the `R40` `dof_torques_l2` and `action_rate_l2` weights equal the upstream values, so we claim no literal "10× regularization" increase. Second, the symmetrized USD fixes a CAD-export asymmetry (a non-zero base i_{xz} and a 1.6×10^{-4} m hip-pitch CoM offset, inducing a $\approx 0.22\,^\circ$/s leftward drift), but "machine-precision symmetry" describes the in-code patch, not a byte-level inspection of the binary asset. Third, the deploy-time action clamp (± 3.0 versus the upstream ± 10000), with a rate limiter, is the safety change motivated by the torque-envelope mismatch of Section 5.

Figure 3 grounds the mechanism: it decomposes the selected policy's per-step reward, showing the recipe-touched terms dominate the penalty side (added `joint_deviation_legs` -0.24 and `dof_torques_l2` -0.14) while velocity tracking (`track_lin_vel_xy_exp` $+0.70$) dominates the positive reward. Lowering the tracking weights and adding these penalties shifts the optimizer's pressure from raw tracking toward joint-posture regularity, the concrete, term-level reason recipe variation produces the cross-engine behavioral differences of Section 6 rather than being incidental.

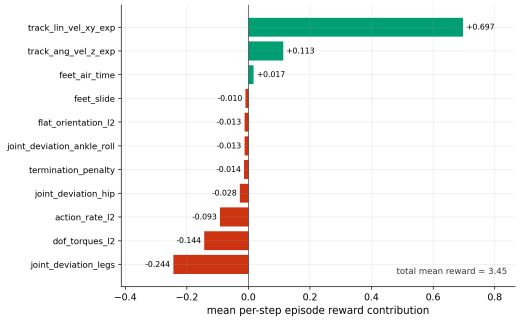


Figure 3. Reward-term decomposition for the selected policy `R40_FullSymUSD` (rank01): mean per-step contribution of each episode-reward component (green = positive, red = penalty), total 3.45. Velocity tracking (`track_lin_vel_xy_exp` $+0.70$) dominates the positive reward; the largest penalties (the recipe-added `joint_deviation_legs` (-0.24) and `dof_torques_l2` (-0.14)) are the terms the ablation actually moves, making the reward-weight diff of Table 1 visual. Components inactive for this run are absent from the source (left out, not zeroed). (source: evidence/figures.md, G6; `runs.csv` row `2026-05-15_22-49-54_R40_FullSymUSD_s42`, `sum/Episode_Reward/*`.)

The 15 staged policies differ only in checkpoint weights, that is, in the training recipe (Table 2): the recipe is the single independent variable. We name `R40_FullSymUSD` (rank01) the *selected policy* throughout, following the cross-engine audit; Section 6 later proposes the statistically tied `R40_V10SymUSD` (rank03) as the *live push-deploy variant* on left/right push-symmetry grounds, so the two are near-interchangeable.

The upstream rank00 baseline functions as the experimental control but was never run through our audit: it has no command-tracking error, tier, cross-engine composite or 73-scenario deep-evaluation block, so its audit cells are "not audited" in Section 6; its one recorded fact is the unclamped ± 10000 action limit.

The final recipe. The converged configuration is: the `R40` trunk (tracking weights $+1.5/+0.75$, `dof_vel_l2` -0.001, `action_l2` -0.001, `joint_deviation_legs`

Table 1. Line-level diff: Berkeley upstream biped configuration versus ours. The unchanged block is marked explicitly. The upstream baseline (rank00) was never run through our audit, so its audit metrics do not exist; the only audited upstream fact is its action clamp. (source: evidence/baseline-diff.md, table (a); upstream paths under Berkeley-Humanoid-Lite/.../config/biped/, our values from POLICIES.md and the deploy YAMLs under policies_top15/.)

Dimension	Berkeley upstream	Our change	Source
Changed			
Linear-vel tracking weight	+2.0	+1.5 (R40)	a
Angular-vel tracking weight	+1.0	+0.75 (R40)	a
dof_vel_l2	absent	added −0.001	a
action_l2	absent	added −0.001	a
joint_deviation_legs	absent	added −0.5	a
USD asset	stock (asym. inertia)	symmetrized family	a
push_robot	commented out	recipe knob (on/off)	a
Fine-tune families	none	AblImit_FT, Gait_FT	a
Action clamp	±10000 (unclamped)	±3.0	a
Unchanged			
PPO / network hyperparams	[256, 128, 128] ELU, ϵ=0.2	unchanged	a
Observation / action space	45-d obs / 12-d act	unchanged	a
PD gains (sim)	k_p=20, k_d=2, 6 N m	unchanged	a
Control rates	25/250/2000 Hz	unchanged	a
Core DR (fric./mass/gains)	startup randomization	unchanged	a
Command ranges (training)	v_x±0.5, ω_z±1.0	unchanged	a

"a" = evidence/baseline-diff.md, table (a). The selected policy is R40_FullSymUSD (rank01), combining the changed rows above with the FullSymUSD asset.

Table 2. Recipe-family glossary: concrete delta of each named family versus the R40 trunk (tracking +1.5/ + 0.75, added regularization, full DR). The selected policy is R40_FullSymUSD (rank01); this glossary is the recipe axis of the controlled ablation. (source: evidence/baseline-diff.md, table (b); policies_top15/README.md lines 7–23 and POLICIES.md.)

Recipe	Concrete delta vs. R40 trunk
R40 trunk	lowered tracking, added reg. terms, full DR (common base)
FullSymUSD	R40 on base+leg inertia-symmetrized USD (**selected**, rank01)
V10SymUSD	FullSym + per-body CoM mirror-averaging (rank03)
AblImit_FT	FT from R40, adds imitation_pose reward (rank02/06/07)
Gait_FT	FT from R40, re-engineered gait rewards; cannot strafe (rank04/05)
R4v2_noPush	DR pushes disabled; longest episodes (439.0 steps) (rank08)
R3v2_noHeight	base-height reward dropped; best MuJoCo trk_fwd, low Isaac (rank15)
V31 family	May-14 standing/balance variants; weaker velocity tracking (rank09–11)

−0.5, full upstream DR with push_robot on) applied to the FullSymUSD inertia-symmetrized asset, trained for 6000 iterations at 4096 parallel environments, deployed with a ±3.0 action clamp. This is R40_FullSymUSD (rank01): the only policy of the campaign to record zero falls across all 73 MuJoCo deep-evaluation scenarios. The R40_V10SymUSD variant (rank03), which adds per-body CoM mirror-averaging to the symmetrized asset (Table 2), is its push-deploy twin: statistically tied on cross-engine robustness and preferred for live push deployment on left-/right push-symmetry grounds (Section 6).

5 Cross-Engine Validation and the Hardware Boundary

The recipe's validation lives inside a deploy stack, but its instrument is purely *cross-engine*: every candidate policy is exported to ONNX and replayed in a second, independent simulator (MuJoCo at 25 Hz, gamepad-driven) and in standalone Isaac Sim, since surviving a simulator the policy was not trained in distinguishes robustness from overfitting (Section 6). All quantitative results reported here are therefore measured in MuJoCo or Isaac Sim. Hardware bring-up reached an end-to-end calibrated robot, with a ROS 2 Jazzy stack driving the motors; a torque-envelope mismatch on hardware motivates the ±3.0 action-clamp row of Table 1. Closed-loop walking on hardware is outside the scope of this report, and the remaining path is sketched in Section 7.

The Isaac Sim interface initially used the OmniGraph ROS bridge, which ticks at render rate (≈6 Hz) against 2000 Hz physics, making an external PD loop unusable. The evaluation configuration therefore delegates PD to PhysX joint drives and publishes position targets only, the same target-only contract as the real deploy stack, so sim2sim and hardware deployment share one code path (source: evidence/sim2real.md).

6 Experiments and Results

This section evaluates the recipe. Holding architecture and deploy configuration fixed (Section 4), we recap the protocol, present the master comparison of recipe variants and what the best variant achieves, show *why* single-engine evaluation would have identified the wrong variant, and close with the by-force engine divergence that bounds the claim. Every number is pinned to its engine (MuJoCo or Isaac Sim) and is a simulation result; none were obtained on hardware (the boundary is in Section 5).

6.1 Evaluation protocol

Each candidate passes through three layers (source: evidence/eval-protocol.md, A.1, A.3, A.5). (i) A **cross-engine gate** of 7 MuJoCo plus 6 Isaac Sim "Core" scenarios; a policy clears Tier-1 only if it falls in none of the 13, places top-third on the Isaac composite composite_S (cutoff ≈0.455–0.458), and reaches MuJoCo walk_fwd tracking ≥ 0.4; 7 of 230 logged runs clear it. (ii) A **MuJoCo-only deep eval** of 73 procedurally generated scenarios (v_x, v_y, ω_z sweeps, mixed-axis, long-drift, step-change, push). (iii) A **push-robustness matrix** of $7 \times 4 \times 4 \times 3 = 336$ scenarios in both MuJoCo and Isaac/PhysX. Two fall criteria stay separate: the RL training termination at base tilt 0.78 rad (≈ 45°), and the looser deep-eval "fell" flag (base-height drop > 0.15 m or pitch/roll > 60°); all "X/73" counts use the latter.

6.2 The master comparison

Table 3 is the controlled comparison: fifteen recipe variants (rank01–15) against the upstream Berkeley reference (rank00, the control). The recipe's best variant, **R40_FullSymUSD (rank01)**, is the unique cross-engine-robust policy: the only one with 0/73 deep-eval falls that also clears the Tier-1 both-engine bar, ranking above the upstream reference on cross-engine robustness within this audit. No single column ranks it first; the cross-engine audit, not any one metric, identifies it.

Read the metrics as competing selection rules. *Best W&B/MuJoCo tracking* (rank15 R3v2_noHeight: err_{xy} 0.204, trk_{fwd} 0.985) has the *lowest* Isaac composite of the tiered set (0.43029) and is not zero-fall (2/73); *best Isaac composite* (rank02 AblImit_FT it6000: 0.50170) logs two deep-eval falls; *longest episode* (rank08 R4v2_noPush: 439 steps, training pushes disabled) is also not zero-fall. Only the combined cross-engine, multi-scenario rule converges on one candidate: **R40_FullSymUSD (rank01) is the unique 0/73 policy**: every other rank logs

Rank	Short name	Grade	err_{xy}	err_{yaw}	ep_len	Falls/73	comp._S	trk_{fwd}	Clamp
00	BerkeleyBipedA	—[a]	—[a]	—[a]	—[a]	—[a]	—[a]	—[a]	±10000
01	**R40_FullSymUSD**	**A**	**0.2213**	**0.8612**	**340.7**	**0/73**	**0.46495**	**0.8835**	±3.0
02	AblImit_FT (it6000)	A	0.2695	1.1771	387.0	2/73	**0.50170**	0.7490	±3.0
03	R40_V10SymUSD	A	0.2178	0.8542	350.9	2/73	0.45859	0.9270	±3.0
04	GaitFinal_FT	—[b]	0.2960	1.2205	391.6	1/73	0.49739	0.7969	±3.0
05	GaitMultiClip_FT	—[b]	0.2960[c]	1.2205[c]	391.6[c]	1/73	0.49021	0.7969	±3.0
06	AblImitSmall_FT	A	0.2315	0.9212	345.7	2/73	0.49448	0.8641	±3.0
07	AblImit_FT (it7498)	A	0.2695	1.1771	387.0	1/73	0.46861	0.9366	±3.0
08	R4v2_noPush	A	0.2257	0.5750	439.0	1/73	0.45499	0.8870	±3.0
09	V31RelStanding	—[b]	0.3619	2.3227	429.5	3/73	0.45648	0.7930	±3.0
10	V31LateralFoot	—[b]	0.3777	2.4679	404.2	3/73	0.43848	0.9619	±3.0
11	V31HipLock	—[b]	0.3633	2.1687	402.7	1/73	0.41347	0.6365	±3.0
12	CleanGaitControl_FT	A	0.2242	0.8766	367.1	1/73	0.44829	0.7983	±3.0
13	R40_SymUSD_FT	A	0.2356	0.9270	356.9	2/73	0.44231	0.8820	±3.0
14	R40_SymUSD	A	0.2244	0.8784	330.7	2/73	0.45320	0.7956	±3.0
15	R3v2_noHeight	S	**0.2044**	0.7973	331.6	2/73	0.43029	**0.9853**	±3.0

Table 3. The master comparison table: Berkeley control (rank00) and the fifteen recipe variants. Columns are candidate selection metrics: err_{xy}/err_{yaw} = command-tracking error; ep_len = `mean_ep_len` (steps); Falls/73 = deep-eval falls (MuJoCo); comp._S = Isaac `composite_S`; trk_{fwd} = MuJoCo `trk_fwd`; Clamp = action limit. Grade = per-run audit tier letter (A/S) from the policy YAML `# tier`, *distinct* from the cross-engine *Tier-1* bar (Section 6.1) that only rank01/02 clear. The selected **R40_FullSymUSD (rank01)** is bold and is the unique 0/73 policy and the cross-engine-robust variant; best err_{xy} and trk_{fwd} both land on rank15, and best comp._S lands on rank02: none of the three single-axis leaders is rank01. [a]rank00 was never audited ("—", not zeros). [b]rank04/05/09/10/11 carry no `# tier` header. [c]rank05's err_vel/trk_{fwd} are copied verbatim from rank04 (header-templating artifact); the two differ on the independent comp._S (0.49021 vs 0.49739). err_vel values are audit-eval (MuJoCo deep eval), not W&B curves. (source: `evidence/results.md`, table (a).)

1–3 falls, each at the out-of-distribution forward speed $v_x = \pm 1.0\,\text{m/s}$ (a couple of V31 at $v_x = +0.7\,\text{m/s}$), outside the trained $v_x = \pm 0.5\,\text{m/s}$ range, and it is the only 0/73 policy that also clears the Tier-1 both-engine bar (rank02 clears Tier-1 but logs 2/73 falls). Per Table 3, rank01 is mid-pack on tracking error, not at either per-axis extreme, while the V31 family (rank09–11) has the worst yaw tracking (2.2–2.5 rad/s).

The control. The Berkeley reference (rank00) was never run through our harness, so its metric cells are "—", not fabricated; its only concrete fact is the unclamped ±10000 action limit versus our ±3.0 deploy clamp (its separate 18/209 deep-eval falls use a different 209-scenario protocol, kept out of the X/73 column). The comparison to Berkeley is therefore a cross-engine robustness result *within this audit*, not a numeric head-to-head benchmark: the upstream reference sits outside the audited Tier-1 set as the unaudited control.

6.3 What the cross-engine validation exposed

A naive ranking would have shipped the wrong policy, and our gate caught it, for a geometric reason. Figure 4 plots MuJoCo forward-tracking against the Isaac composite per policy; the two April cluster runs that topped the W&B curves occupy the high-MuJoCo, low-Isaac corner the Tier-1 policies vacate. The numbers quantify it: `bigger_envs` has the best W&B err_vel_xy of any biped (0.194, MuJoCo trk_{fwd} 0.96, 0 gate-falls) yet the lowest Isaac composite of the batch (0.305); the `2026-04-15` run (err_{xy} 0.207, trk_{fwd} 0.89, 0 gate-falls) reaches only 0.356; and `relax_reg_stack` fell in MuJoCo (trk_{fwd} 0.70, 1 fall) while scoring 0.15 in Isaac. The two Tier-1 policies trade some MuJoCo tracking for a higher Isaac composite (rank01 0.88 → 0.465, rank02 0.75 → 0.502, both 0 gate-falls), showing the regions the gate separates (April-run "falls" are 0-fall gate counts, not the X/73 deep eval; source: `evidence/eval-protocol.md` A.4, `cross_engine.md` L36–47).

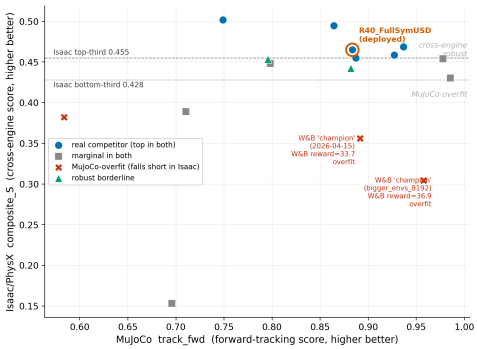


Figure 4. Cross-engine generalization: MuJoCo forward-tracking (`track_fwd`, x) versus Isaac/PhysX composite (`composite_S`, y) per policy, colored by audit verdict. The two old W&B high-reward runs sit far right (MuJoCo track_fwd 0.89/0.96) yet lowest on the Isaac axis (0.36/0.30), the overfitting that motivates the gate; the selected R40_FullSymUSD is ringed. Their annotated W&B mean rewards are 36.9 (`bigger_envs_s42_8192`) and 33.7 (`2026-04-15` run); MuJoCo-only Tier-3 rows are omitted, not zero-filled. (source: `evidence/figures.md`, G3; `cross_engine.csv`; thresholds from `cross_engine.md`.)

Why one checkpoint and one metric mislead: the peak-versus-final regression. A sharper instance comes from one run, whose err history is in Figure 2 panel (b). Ranks 02 and 07 are *the same run* (W&B id `aqe6jlq5`) read at two checkpoints: err_xy peaks at 0.1976 (step 6013) then drifts to 0.2916 (step 7488), a +48% regression. The hazard compounds across measurement systems: the *audit-eval* err_xy for this run is 0.2695 (`cross_engine.csv`) while the *W&B curve* gives peak 0.198/final 0.292; different harnesses on different scenario sets, so a number is meaningful only paired with *which checkpoint, which metric, and which harness* produced it (source: `evidence/results.md`, (c); `peak_vs_final.csv`).

6.4 Push robustness and the by-force engine divergence

We stress-tested the top three policies over the 336-push matrix (0.1 s pulse after motion stabilizes; binary fall/no-fall). In MuJoCo, push survival is 90.5%: R40_FullSymUSD (rank01) and R40_V10SymUSD (rank03) both log 32/336 falls and AblImit_FT at iter6000 logs 33/336 (a three-way tie within one fall). The *live push-deploy variant* R40_V10SymUSD (rank03) was selected on its symmetric L/R push survival (0.0 pp direction delta versus 1.2 pp for the FullSym variant), not on survival count (source: `evidence/results.md`, tables (b.1, b.5)).

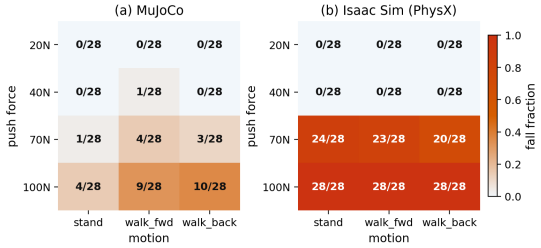


Figure 5. Push-robustness heatmap for **R40_FullSymUSD (rank01)**: fall fraction per (force, motion) cell across MuJoCo (a) and Isaac Sim/PhysX (b), 28 scenarios per cell (336 total). The engines agree at $\leq$ 40 N (0–1 falls in both), then diverge qualitatively at 70 N (MuJoCo 8/84 falls, 1+4+3 across stand/walk_fwd/walk_back, versus Isaac 67/84, 24+23+20) and separate completely at 100 N: MuJoCo 23/84 falls (4+9+10) versus complete failure in Isaac (84/84 across all three motions). This cross-engine instrument detects systematic engine-level differences in contact solving and integration whose precise attribution we leave open, a divergence invisible to MuJoCo-only evaluation. (source: `logs/robustness/2026-05-18_top3` and `isaac_2026-05-18` per-scenario JSONs; totals 32/336 MuJoCo, 151/336 Isaac.)

The cross-engine instrument matters most here. The 90.5% figure is MuJoCo-only; the identical matrix in Isaac/PhysX gives only ≈55% for the deploy variant: R40_V10SymUSD (rank03) survives 185/336 (= 55.1%, a −35.4 pp gap) with MuJoCo by-force counts of 0/84 (20 N), 1/84 (40 N), 9/84 (70 N), and 22/84 (100 N). Rank01 records the same 151/336 Isaac falls (0+0+67+84, the caption totals of Figure 5), so the two are indistinguishable on aggregate Isaac survival. Figure 5 (rank01) shows the engines agree at $\leq$ 40 N, then diverge qualitatively at 70 N and separate completely at 100 N (Isaac 84/84 versus MuJoCo 23/84), so a policy selected on MuJoCo alone would be blind to that 70 N cliff. Both deployed policies tie at 32/336 MuJoCo falls. The strict engine-agnostic claim (0 falls in *both* engines) is therefore the $\leq$ 20 N zero-fall envelope; the $\leq$ 40 N range is operator-safe (a single MuJoCo fall, 0 Isaac), and the 90.5% MuJoCo figure is engine-specific (source: `evidence/results.md`, (b.3, b.6); `evidence/eval-protocol.md`, A.5).

Within MuJoCo, the force × motion structure of Figure 5 is benign at 20 N (0/28 across every motion) and worst at 100 N during walking (9–10/28 for rank01), with rank01 leading every behavioral category except idle standing (largest margin on push recovery, 0.715 versus 0.305 for the 25hza reference; source: `evidence/figures.md`, screenshot 4).

7 Discussion and Future Work

7.1 What the recipe needed

Three ingredients carried the result. The symmetrized asset family is the decisive axis, separating the zero-fall `R40_FullSymUSD` (rank01) and its sibling (rank03) from the campaign trunk by removing the 0.22°/s CAD-asymmetry drift; the motion-imitation objective is the second, its phase-indexed pose reward (Section 3.4) producing rank02, the best Isaac composite (0.502) of the audited set; both sit on the trunk reward shape (lowered tracking weights, three added regularization terms). On this controlled set, `R40_FullSymUSD` is the unique 0/73 policy and one of seven of 230 runs to clear the both-engine Tier-1 bar, the only variant to do both. Every number here is a *simulation* number (MuJoCo, or cross-engine MuJoCo plus Isaac Sim); the physical robot is calibrated but closed-loop walking is not yet demonstrated.

The validation lesson is the second movement of the same story: *no single metric selects a cross-engine-generalizing policy, and our cross-engine multi-scenario audit does.* Best training/MuJoCo tracking, best Isaac composite, and longest episode each would have shipped a different, non-zero-fall policy (Section 6); only the combined audit isolates `R40_FullSymUSD`, rejecting the high-reward April runs that top the W&B curves yet land bottom-two in Isaac. We thus treat the second engine as a held-out test distribution. The same discipline bounds the robustness claim: the 90.5% MuJoCo push survival drops to 55.1% in Isaac (−35.4 pp), so the defensible engine-agnostic claim is the $\leq$ 20 N strict zero-fall envelope (0 in *both* engines), with $\leq$ 40 N operator-safe (1/84 MuJoCo, 0 Isaac); on the deep eval every other rank falls only at out-of-distribution speeds (beyond ±0.5 m/s), justifying the anti-OOD deploy envelope.

7.2 What did not work

Several explorations beyond the ingredients that made the recipe did not reach deployment, each clarifying a design choice. (i) *Adversarial imitation.* We implemented an AMP [12] discriminator and reference-motion pipeline from scratch over RSL-RL (no native AMP support), self-auditing it to a green integration smoke test; of five cluster runs launched, none was carried to completion within scope, so the AMP path is validated to smoke test, not trained to convergence, and the phase-indexed pose reward (Section 3.4) is the imitation route that reached the audited set. (ii) *Warm-start transfer from a raised-shoulder variant.* Seeding the biped from a taller `MinaRaised` platform via `--resume` decayed back to the from-scratch attractor within roughly a thousand iterations in all sixteen attempts: the inherited value function stays calibrated to the source-task reward and biases the PPO advantages toward the prior behavior, so naive checkpoint reuse across reward definitions fails. The taller platform compounded it: its higher centre of mass (0.517 versus 0.484 m) roughly halved the random-init exploration window (1.2 → 0.4 s), so from-scratch training on it failed too. (iii) *A gait-phase observation.* A sin/cos gait-phase clock was explored but not kept: the deployed biped retains the upstream 45-d observation, trading a possible gait prior for a smaller deploy-time gap. (iv) *Re-engineered gait shaping.* The `Gait_FT` family (rank04/05)

tracks competitively within its branch but *cannot strafe* (its lateral velocity command is hard-zeroed), so it is excluded from the deployable set. (v) *The MuJoCo-overfit high-reward runs* (Section 6.3) are the central negative result: the highest-reward policies do not survive the second engine, the failure mode the cross-engine audit exists to catch.

7.3 Hardware boundary and outlook

On hardware the robot is calibrated end-to-end (22/22 joint mapping, leg residuals broadly within $\sim 2°$) and the deploy/safety stack is real, exercised, and fault-free, but closed-loop walking is not achieved: the real `rl_controller` was never launched and the sim's MuJoCo policy drove the motors off sim state. Closing that gap (the real controller, the gamepad, the $k_p = 20$ sim versus 50 real PD mismatch) is left as future work; the Berkeley public demo, on an unreleased concurrent-state-estimation architecture (a wider $[256, 256, 256]$ actor) we could not match against the public $[256, 128, 128]$ checkpoint, was likewise out of scope. For cost context, the campaign comprised 230 logged runs over eight weeks ($\sim$268 h logged training wall-clock, $\sim$97 h on the biped stream) on EPFL SCITAS Kuma L40S nodes and a local workstation. The twofold contribution (a robustness-hardened biped policy and the cross-engine audit that selected it) is recapped in Section 8.

8 Conclusion

We developed, through a 230-run, eight-week training campaign on the Berkeley Humanoid Lite biped [6], a PPO training recipe: lowered tracking weights with three added regularization terms, a left/right-symmetrized asset family, a push-randomization knob, and a DeepMimic-style motion-imitation objective. Holding architecture and deploy configuration byte-identical across a controlled 15-policy comparison and varying only this recipe, its best variant is the unique 0/73-fall policy of the campaign, `R40_FullSymUSD`, one of seven of 230 runs to clear the both-engine Tier-1 bar, ranking above the unaudited upstream Berkeley reference on our cross-engine robustness tiers. Validating candidates in a second physics engine proved essential, since best training reward, best MuJoCo tracking, and best episode length each pick a different, non-zero-fall policy. The recipe's best variant survives 90.5% of MuJoCo pushes and 55.1% in Isaac; the strict zero-fall claim holds to the $\leq$ 20 N envelope, and the $\leq$ 40 N range stays operator-safe (0 Isaac falls, 1/84 MuJoCo). The portable contributions are thus the recipe and its cross-engine, multi-scenario validation protocol, not any single trained policy. The hardware boundary stands: the robot is calibrated and the ROS 2 Jazzy stack drives its motors, but closed-loop walking is the next milestone.

References

[1] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov. Proximal Policy Optimization Algorithms. arXiv:1707.06347, 2017.

[2] K. P. Murphy. Reinforcement Learning: An Overview. arXiv:2412.05265, 2025.

[3] J. Tobin, R. Fong, A. Ray, J. Schneider, W. Zaremba, and P. Abbeel. Domain Randomization for Transferring Deep Neural Networks from Simulation to the Real World. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2017. arXiv:1703.06907.

[4] A. Kumar, Z. Fu, D. Pathak, and J. Malik. RMA: Rapid Motor Adaptation for Legged Robots. In *Robotics: Science and Systems (RSS)*, 2021. arXiv:2107.04034.

[5] I. Radosavovic, T. Xiao, B. Zhang, T. Darrell, J. Malik, and K. Sreenath. Real-world humanoid locomotion with reinforcement learning. *Science Robotics*, 9(89):eadi9579, 2024.

[6] Y. Chi, Q. Liao, J. Long, X. Huang, S. Shao, B. Nikolic, Z. Li, and K. Sreenath. Demonstrating Berkeley Humanoid Lite: An Open-source, Accessible, and Customizable 3D-printed Humanoid Robot. In *Robotics: Science and Systems (RSS)*, 2025. arXiv:2504.17249.

[7] M. Mittal, C. Yu, Q. Yu, J. Liu, N. Rudin, D. Hoeller, J. L. Yuan, R. Singh, Y. Guo, H. Mazhar, A. Mandlekar, B. Babich, G. State, M. Hutter, and A. Garg. Orbit: A Unified Simulation Framework for Interactive Robot Learning Environments. *IEEE Robotics and Automation Letters*, 8(6):3740–3747, 2023. arXiv:2301.04195.

[8] N. Rudin, D. Hoeller, P. Reist, and M. Hutter. Learning to Walk in Minutes Using Massively Parallel Deep Reinforcement Learning. In *Conference on Robot Learning (CoRL)*, pp. 91–100, 2022. arXiv:2109.11978.

[9] Y. Seo, C. Sferrazza, J. Chen, G. Shi, R. Duan, and P. Abbeel. Learning Sim-to-Real Humanoid Locomotion in 15 Minutes. arXiv:2512.01996, 2025.

[10] Z. Gu, J. Li, W. Shen, W. Yu, Z. Xie, S. McCrory, X. Cheng, A. Shamsah, R. Griffin, C. K. Liu, A. Kheddar, X. B. Peng, Y. Zhu, G. Shi, Q. Nguyen, G. Cheng, H. Gao, and Y. Zhao. Humanoid Locomotion and Manipulation: Current Progress and Challenges in Control, Planning, and Learning. arXiv:2501.02116, 2025.

[11] X. B. Peng, P. Abbeel, S. Levine, and M. van de Panne. DeepMimic: Example-Guided Deep Reinforcement Learning of Physics-Based Character Skills. *ACM Transactions on Graphics (SIGGRAPH)*, 37(4):143, 2018. arXiv:1804.02717.

[12] X. B. Peng, Z. Ma, P. Abbeel, S. Levine, and A. Kanazawa. AMP: Adversarial Motion Priors for Stylized Physics-Based Character Control. *ACM Transactions on Graphics (SIGGRAPH)*, 40(4):144, 2021. arXiv:2104.02180.